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# Tissue AI

A visual tool that automatically transforms your text into customizable illustrations

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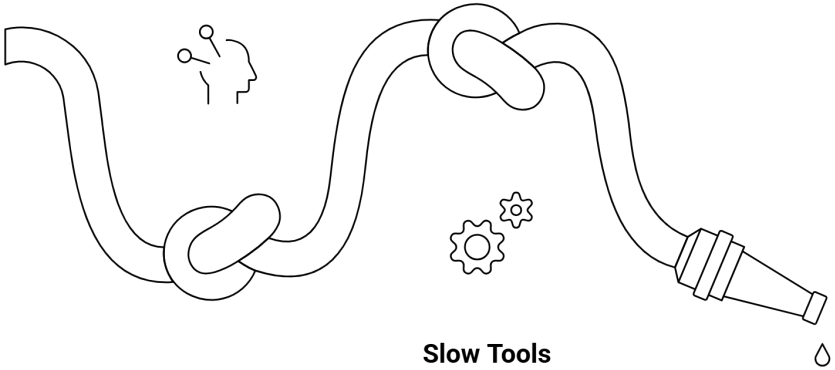
# Problem Definition

**Manual Creation**

Time-consuming and mentally taxing



**Slow Tools**

Planning hinders clear thinking

→ Digital information grows very fast.

→ Users struggle to read, organize, and understand large text content.

→ Lecture notes, ideas, documents, and workflows become hard to analyze using only text.

→ Visual forms like mind maps and flowcharts improve understanding.

**But the issue is**

# Objective

The goal is to design and develop Tissue AI.

Tissue AI is an intelligent system that converts text into visual diagrams. The system creates flowcharts and mind maps from written content.

The system reduces manual effort during diagram creation. It improves knowledge visualization using Gen AI techniques.

# Methodology

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follows a **Generative AI-driven text-to-visualization pipeline.**

**transforms unstructured natural language input into structured visual representations**

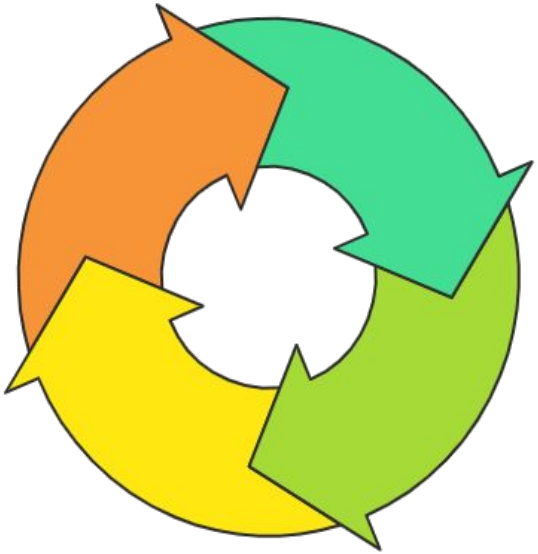
- **Methodological Approach**
  - Generative AI
  - NLP Techniques
  - Structured Intermediate Representation
- **Tissue AI**
- **Data Analysis Procedures**
  - Text Preprocessing
  - Semantic Understanding
  - Structure Generation
  - Tools and Technologies
- **Data Collection Techniques**
  - User-Provided Text
  - On-Demand Collection

does not rely on traditional datasets such as surveys or labeled training samples. Instead, it operates on **dynamic user-provided textual input.**

# Project execution & QA

1. System design and GenAI model selection
2. Text preprocessing and prompt structuring
3. Generative relationship and structure generation
4. Diagram rendering using visualization frameworks
5. Testing and iterative refinement

## Quality Assurance Cycle



**Test Consistency**

Ensure consistency across input types

**Validate Logic**

Ensure diagram logic is sound

**Gather Feedback**

Collect user feedback for refinement

**Check Clarity**

Verify visual clarity and readability

# INNOVATION

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POINT 01

Generative AI to automatically convert unstructured text into structured visual representations

POINT 02

The system eliminates user-defined layouts by dynamically generating flowcharts and mind maps through AI-driven semantic understanding.

POINT 03

This enables faster learning, scalable understanding, and forms a foundation for future AI-assisted learning and visualization systems.

# Software Requirements

| Development | Large Language Models | Testing and Deployment |
| --- | --- | --- |
| Python | OpenAI API | PyTest |
| JavaScript | Google Gemini API | Postman |
| React.js | Anthropic Claude API | Docker |
| Visual Studio Code | Open-source LLM APIs | GitHub Actions |
| Git | | Vercel |
| GitHub | | |